

Example: اداره → ادار

Example: گوبنه → گوبن

Example: لومړي → لومړ

Example: رسېدل → رس

These seven letters are the commonest single-letter endings in Pashto: the five ye letters mark gender, number and person, ه marks the feminine and adjectival agreement, and ل is the infinitive ending. On a short word a final letter from this set is nearly always an ending rather than part of the root.

The upper bound of five letters is the part to question. Beyond it the rule starts cutting roots: held-out accuracy falls from 59.2% at five letters to 58.2% at six and 54.0% with no bound. Longer words are left to Rules 6 to 8, which know which particular suffix they are removing.

Rule 5: The word ends in -ستان and at least three letters remain

→ remove -ستان and stop; no further rule is applied

Affixes (1): ستان

Example: افغانستان → افغان

Example: بلوچستان → بلوچ

Example: ترکمنستان → ترکمن

Example: هندوستان → هندو

-ستان forms place names from a noun: ستان + بلوچ, ستان + افغان. It is an ordinary derivational suffix and is treated as one; a place name gets no special protection.

The rule stops rather than continuing because the general rules would keep going: ترکمنستان first loses ستان to give ترکمن, and then Rule 7 would take the -من and leave ترک. Stopping is the whole point of stating it separately.

Part 2 — Rules that propose an analysis

These do not decide anything on their own. Each proposes a candidate stem; they may apply on top of one another up to three times; and Rule 10 chooses between the results. A single-pass stemmer must commit to its first guess, which is the main structural difference from the baseline.

In the tables below, 'example word' is a real word from the project's annotated data and 'stem produced' is what the running code returns for it. Nothing in these columns was typed by hand.

Rule 6: The word ends in an inflectional suffix and at least two letters remain

→ propose the word without that suffix

Inflection changes the form of a word without changing the word itself: **کور** and **کورونه** are the same noun, singular and plural. These are the longer inflectional endings, so two remaining letters is enough — there is little chance of confusing **ونو-** with root material.

The **پر-** and **بد-** group belongs here too. These form verbs from nouns and adjectives by a regular pattern (**جوړ** → **جوړېږي** , **جوړېدل** , **جوړېږي**), which is why those verbs are not in the dictionary.

Affix	Min stem	Example word	Stem produced	What it marks
کانو	2	ماماګانو	ماما	oblique plural ګانو-
ګانې	2	ویدیوګانې	ویدی	feminine plural of vowel-final stems
ګانې	2	ویدیوګانې	ویدی	direct plural ګانې-
ګونه	2	سلګونه	سل	
ګونو	2	زرګونو	زر	
انو	2	روانو	روان	oblique plural انو-
ونه	2	حقونه	حقون	direct plural
ونو	2	رګونو	رګ	oblique plural
ګان	2	خپګان	خپ	plural -gān
ېدا	2	—	—	derivative verbal noun
ېدل	2	رسېدل	رس	inchoative infinitive -edəl
ېده	2	ورسېده	ورس	eda
ېدو	2	کمېدو	کم	derivative oblique infinitive
ېږم	2	ودرېږم	ودر	derivative present 1sg
ېږو	2	پوهېږو	پوه	derivative present 1pl
ېږي	2	رسېږي	رس	derivative present: ،
لو	2	کولو	کول	oblique infinitive
ول	2	اصول	اصل	causative -awəl
ېد	2	اوسېد	اوس	derivative past stem: ،
ېږ	2	—	—	derivative present stem

ل	2	اصول	اصل	NOT stripped. The verbal -ل is real, but a sta...
ې	2	يوې	يو	feminine / plural -ې

Rule 7: The word ends in a short inflectional suffix and at least three letters remain

→ propose the word without that suffix

The same kind of ending, but only one or two letters long. A longer minimum stem is required because a short ending is much harder to tell from the end of a root: the ۀ of لنډه is a suffix, the ۀ of سيمه is part of the word.

Affix	Min stem	Example word	Stem produced	What it marks
تانه	3	راستانه	راس	pl of -tun
تانو	3	ويښتانو	ويښ	obl pl of -tun
اتو	3	مهماتو	مهم	Ar. pl + obl
انې	3	سفارتخانې	سفارتخان	alt. direct plural
جات	3	—	—	Pers. pl -jāt
ات	3	جومات	جوم	Ar. sound plural
ان	3	رحمان	رحم	animate plural -ان
ون	3	بدلون	بدل	adj->noun: ،
وو	3	خواوو	خوا	F3 obl. pl
وې	3	ارتياوې	ار	F3 dir. pl
يو	3	رسنيو	رسنى	oblique plural: کورنيو، دوستيو
ئ	3	وکرئ	وکر	imperative pl -əy
ه	3	لرله	لرل	fem/agreement -a; never removed if <3 chars wo...
و	3	يادو	ياد	oblique plural -و
ى	3	مرمى	مرم	fem sg -əi (often lexical!)

Rule 8: The word ends in a derivational suffix and at least three letters remain

→ propose the word without that suffix

Derivation makes a new word rather than a new form: کار work gives کارگر worker, چټک fast gives چټکوالی speed. Strictly these are different lexemes, so removing them is a decision rather than a fact, and it is taken here because the purpose is to group related words together.

Affix	Min stem	Example word	Stem produced	What it marks
يدونکي	3	اوسيدونکي	اوس	derivative agentive, ى spelling
بدونکي	3	زياتبدونکي	زيات	derivative agentive

خانه	3	—	—	place
داني	3	—	—	container (Persian)
ستان	3	پاکستان	پاک	Persian place suffix. NOT stripped: the gold s...
غالی	3	لوبغالی	لوبغا	DO NOT STRIP: one attested example (لوبغالی); ...
والو	3	نړيوالو	نړی	oblique of -wāli
والی	3	نشتوالی	نشت	adj->noun abstract
ونکو	3	خپروونکو	خېر	oblique agentive
ونکی	3	خپروونکی	خېر	agentive participle -unkay: دراوونکی
ونکې	3	کېدونکې	کېد	fem agentive
کوټی	3	—	—	needs validation: no attested example
گلوې	3	—	—	relationship
گوټی	3	ښارگوټی	ښارگ	needs validation: no attested example
انه	3	ارزانه	ارز	manner, -like/-ly: ماشومانه، زېږورانه، دوستانه
اوی	3	درناوی	درنا	GATED by the gold standard: پوهاوی، سپکاوی are...
بان	3	طالبان	طالب	keeper/guard (Persian): دربان، باغبان
توب	3	غړیتوب	غړی	adj->noun abstract
تون	3	روغتون	روغ	adj->noun: ،
تیا	2	پکتیا	پکت	abstract
خور	3	—	—	consumer
دار	3	—	—	possessive
دان	4	—	—	container (Persian)
زار	3	—	—	place of abundance (Persian)
لیک	3	برخلیک	برخ	compounding -lik: ،
مند	3	هنرمند	هنر	possessive -mand: ،
ناک	3	خطرناک	خطر	adj
نکو	3	ښوونکو	ښوو	obl of -unkay
نکی	3	ښوونکی	ښوو	agentive/present participle -unkay: ،
نکې	3	کېدونکې	کېد	fem of -unkay
وال	3	نړۍ وال	وال	GATED by the gold

				standard: نړۍ وال is kept who...
وان	3	گريوان	گری	needs validation: the earlier example دربان be...
ولي	3	—	—	needs validation: no confirmed attested exampl...
وتی	3	ښارگوټی	ښارگ	diminutive: ،
وکی	3	—	—	DO NOT STRIP: lexicalized (وړوکی)
پال	3	—	—	keeper/supporter, noun AND adjective
پوه	3	کارپوه	کار	expert in: ،
ځای	3	—	—	place: اوسېدنځای
ځيو	3	ښوونځيو	ښوون	obl of -dzay
چکی	3	—	—	diminutive (adj)
ښنه	3	اندېښنه	اندېښ	action
کار	3	اهلکار	اهل	agentive
گار	3	ټينگار	ټين	agent, distinct from کار-
گره	3	—	—	adj->noun
گری	3	ځانگری	ځان	agentive
يال	3	—	—	GATED by the gold standard: مرستیال، ar... خبريال
یزه	3	لمریزه	لمر	feminine of یز-
یزو	3	کرنیزو	کرن	oblique -izo
ار	4	اهلکار	اهل	GATED by the gold standard: میانمار، ar... ټينگار
اک	4	خطرناک	خطر	action/product: خوراک، څښاک، پوښاک
جن	3	—	—	characterized by
لې	3	فعالې	فعال	fem past participle
من	3	پرچمن	پرچ	adj
نه	3	حقونه	حقون	verb->noun
نی	3	ورځنی	ورځن	NOT stripped as a unit. The gold removes only ...
نۍ	3	ورځنۍ	ورځن	NOT stripped, for the same reason as نۍ-: rule...

نې	3	ساتنې	ساتن	NOT stripped. -نې، -نی are one morphem...
ور	3	—	—	GATED by the gold standard: پېښور، مشهور، لاهو
ور	3	باجور	باج	able/worthy
ين	3	—	—	GATED by the gold standard: ناوړين، واکسين، پو
ځی	3	پوهنځی	پوهن	place
چي	3	—	—	DO NOT STRIP: Turkic, lexicalized (موچي)
ښت	3	بندښت	بند	adj->noun
کر	3	کارگر	کار	agentive -gar
گی	3	تولگی	تولگ	diminutive: ،
یت	3	ملکیت	ملک	Arabic abstract-noun -iyyat: ، ،
یز	3	ښونیز	ښون	relational -iz: ولسمشریز
ا	4	پکتیا	پکت	abstract from adjective (). NOT stripped: a si...
ت	4	ملکیت	ملک	GATED by the gold standard: حکومت، وزارت، سفار
ک	4	—	—	GATED by the gold standard: ترڅنگ and similar ...
ی	3	هندي	هند	abstract/nisba: ،

Rule 9: The word begins with a prefix, at least three letters remain, and what remains is an attested word

→ propose the word without that prefix

Pashto has few productive prefixes. نا- and بې- negate, هم- means 'co-', لا- intensifies. The attestation condition is what makes the rule safe: بېجینگ (Beijing) begins with بې but جینگ is not a Pashto word, so nothing is removed.

The verbal particles وا، پرې-، وړ-، در-، را-، و- were implemented and then removed. Measured on the annotated data they stripped correctly between 13% and 50% of the time, because a word-initial و or را cannot be told apart from the first letter of a root.

Affix	Min stem	Example word	Stem produced	What it marks
بلا	3	—	—	without (Arabic, restricted)
بیا	3	بیارغونه	بیارغ	repetition
غیر	3	غیرقانونی	غیرقانون	KEPT, not stripped: the gold correction log re...
نیم	3	نیمایی	نیمای	half

بې	3	بېرته	بېرت	KEPT, not stripped: the gold correction log re...
سر	4	سرېږه	سرېږ	head/chief (ر). NOT stripped: سر is a free noun...
ضد	3	—	—	anti- (Arabic); often written separately: ضد ا...
لا	3	لاسته	لاست	privative in Arabic learned formations: لالانته...
نا	3	نامتو	نامت	KEPT, not stripped: the gold correction log re...
هم	3	همغږې	همغږ	co-/fellow

Part 3 — Choosing between the proposals

Rule 10: Several rules proposed different stems

→ score each proposal and return the highest

The score adds: how much affix material was removed; a bonus if other words in the corpus are built on the same stem; a bonus if the stem itself occurs. Being attested raises a proposal's score but never blocks one, because a stem does not have to be a dictionary word. The unstripped word receives only a small attestation bonus, since the input is almost always in the corpus and that says nothing about whether it is a stem.

This is where the system differs most from a single-pass stemmer. For کورونو the proposals are کورونو unchanged, کور by Rule 6, and کورون by Rule 7. All three are legal; the score decides. A first-match stemmer has to take whichever rule it happens to try first.

Rule 11: A proposal would leave fewer letters than the rule's minimum

→ discard that proposal

Without this کېدل reduces to ک.

Each rule carries its own minimum rather than one global threshold, because a three-letter suffix can safely leave less behind than a one-letter suffix can.

Rule 12: A rule marked first-pass-only would apply to a word that has already lost material

→ do not apply it

The verbal -ل must not come off an adjective root that happens to end in بېنکل → بېنکلې: ل is right, بېنکل → بېنکلې is not.

Some endings are only endings on an intact word. Once material has already come off, what now sits at the end of the word may be root.

A rule that was tested and removed

One further rule was proposed, implemented and then deleted: remove the last three letters of any non-Pashto word of seven letters or more. It is recorded here because a rule that was tested and rejected is part of the method.

Judged against	Cut three letters	Remove a documented affix
Development set, 2,912 words	25.0%	45.0%
Held-out set, first annotation	59.7%	35.5%
Held-out set, re-annotated	43.5%	51.6%

Removing a real affix wins on both careful annotations. Every word the rule damaged carried a genuine affix:

درلو → درلودلو where the affix is -مسو- where it is -لو, مسوولیت → مسو- where it is -لو, اعلانات → اعلان- where it is -ات. The rule fired before the affix list was consulted and cut through the root.

The irregular-verb dictionary used by Rule 2

34 verbs checked against a printed grammar:

Infinitive	Present stem	Past stem	Meaning
کول	کو	کړ / کاوه	to do/make
کېدل	کېږ / کېږي	شو / شول	to become/happen
تلل	ځ	لار / تل	to go
راتلل	راځ	راغل / راغی	to come
وړل	وړ	یوړ / یوړي	to carry/take away
راوړل	راوړ	راوړي	to bring
خوړل	خور	خوړ	to eat
لیدل	وین	لید	to see
کتل	گور	کوت / کت	to look
اخیستل	اخل	اخیست	to take/buy
نیول	نیس	نیو	to catch/hold
ویل	وای	ویل	to say
ورکول	ورکو	ورکړ	to give
غوښتل	غواړ	غوښت	to want
موندل	موم	موند	to find
لوستل	لول	لوست	to read
وژل	وژن	وژل / ووژه	to kill
پېژندل	پېژن / پېژني	پېژند / پېژند	to know/recognize
ښودل	ښی	ښود	to show
الوتل	الوز	الوت	to fly
ختل	خېژ	خوت / خت	to climb/rise

بلل	بول	بلل	to call/consider
ایستل	باس	ایست / وېست	to pull out/extract
اغوستل	اغوند	اغوست	to wear
اوړل	اوړ	وېښت / اوښت	to pass/cross
چاودل	چو	چاود	to explode
نڅېښتل	نغاړ	نڅېښت	to wrap/roll up
وېشتل	ول	وېشت	to shoot
وتل	وځ	وت	to go out
رودل	رو	رود	to suck
ورتل	ورځ	ورتل / ورغل	to go there
درتل	درځ	درتل / درغل	to go to you
راکول	راکو	راکړ	to give here
درکول	درکو	درکړ	to give to you

8 verbs not yet checked against a printed grammar:

Infinitive	Present stem	Past stem	Meaning
لرل	لر	درلود / لرل	to have
اورېدل	اور	اورېد	to hear
کېښودل	ږد	کېښود / ایښود	to put/place
پرېښودل	پرېږد	پرېښود	to leave/abandon
پرېوتل	پرېوځ	پرېوت	to fall
کښېښتل	کښېن	کښېناست / ناست	to sit down
گڼل	گڼ	گاڼه / گڼل	to count/consider
پرانیستل	پرانیږ	پرانیست	to open